MIRAGE: Optimal, Road-Graph-Native Location Obfuscation at City Scale for IoT Smart Cities

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Abstract—IoT smart cities depend on continuous location reporting, and a large family of obfuscation mechanisms, including k-anonymity, differential privacy (DP), graph constraints, and temporal cloaking, has been proposed to protect it. Two problems remain open in a deployable, quantitative form. The first is the absence of a common, threat-aware scale on which the privacy of different mechanisms can be compared. The second is the unknown distance between the heuristics that are actually deployed and the best privacy achievable on a real road network. This paper addresses both. We adopt an adversary-grounded privacy metric, the expected error in metres of an optimal Bayesian adversary, and instantiate it under two threat models: a snapshot adversary and a causal Bayesian filtering trajectory adversary equipped with a Markov mobility prior. The two adversaries rank mechanisms differently, which shows that a single privacy scale is inadequate. We then introduce MIRAGE, a road-graph-native optimal obfuscation mechanism that scales to a real city. The Bayesian-optimal mechanism is the solution of a linear program (LP) that maximises the adversary's error subject to a utility budget, with both distortion and adversary error measured as graph shortest-path distance. MIRAGE makes this tractable by solving one small LP per density-adaptive local region, which turns an intractable $|V|^2$ program into $|V|$ tiny ones; the resulting mechanism is locally optimal and, we show empirically, within 1.5% of the global optimum in the operationally relevant regime. On real OpenStreetMap networks of two cities under three mobility datasets (GeoLife and T-Drive in Beijing; a taxi dataset in Porto) we measure the price of heuristics: at matched utility, DP and k-anonymity forfeit 5–22% of the achievable privacy in the operationally relevant regime, whereas MIRAGE tracks the global privacy–utility frontier up to the network's intrinsic uncertainty ceiling. The cross-city variation is consistent with prior heterogeneity, the structure that the optimal mechanism exploits and the heuristics ignore. Finally, we report a negative result that snapshot-optimality does not imply trajectory-optimality: MIRAGE's scalability partition leaks region membership, so against a sequential adversary a diffuse mechanism can be more robust. This identifies the central open problem. The complete framework is released for reproducibility.

Index Terms—Location privacy, optimal obfuscation, Bayesian adversary, differential privacy, k-anonymity, road networks, trajectory privacy, linear programming, Internet of Things, smart cities.

1 Introduction

Every connected vehicle that reports its position every few seconds traces out its driver's home, workplace, clinic visits, and daily routine, and stripping the device identifier hides little of this [1], [2]. Smart-city deployments produce exactly these streams at scale, for analytics, traffic management, and emergency response, which is why location obfuscation has been studied for two decades. The mechanisms that resulted span spatial k-anonymity [1], [2], [10], differential privacy (DP) and geo-indistinguishability [3], [8], graph-constrained perturbation [7], density-aware adaptation [12], and temporal cloaking [2].

A practitioner choosing among them for a real road network still confronts two problems that the literature does not resolve in a deployable, quantitative way.

Comparable, threat-aware privacy. Mechanisms are scored on incompatible, family-specific scales: an anonymity parameter k, a log-scaled privacy budget ε, a cloaking-region area, or an information-loss figure, none of which carries a common operational meaning. A cloaking region of 500 m radius and a Laplace noise of 500 m standard deviation do not provide the same privacy. Mechanisms are also almost always evaluated against a single snapshot attacker, including mechanisms such as temporal cloaking whose purpose is to defeat trajectory tracking. Without a common, threat-aware scale, the question of which mechanism is more private has no defensible answer.

Distance from the optimum. Classical theory shows that, for a given utility budget, there exists an optimal obfuscation mechanism, the solution of a linear program (LP), that maximises a Bayesian adversary's expected error [5]–[7]. That theory has only ever been instantiated on the Euclidean plane or small grids, is snapshot-focused, and does not scale, because the LP has $|V|^2$ decision variables. It has therefore never been evaluated on a real road network, and the gap between the heuristics people deploy and the optimum on real topology is unknown.

A motivating scenario. A city runs 10,000 connected vehicles reporting every 10 s. The operator can spend, say, 500 m of average spatial distortion and still keep analytics useful. The operator needs the release policy that buys the most privacy for that budget on this city's roads and mobility. Candidate answers include DP with a tuned ε and k-anonymity with a tuned k; a provably better answer is not available today, and neither is a measurement of how much privacy a suboptimal choice forfeits.

This paper resolves both problems and quantifies the scenario.

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Contributions.

1) An adversary-grounded, threat-model-aware privacy metric (Section 5). Privacy is the expected error of an optimal Bayesian adversary [4], in metres, evaluated under both a snapshot adversary and a causal Bayesian filtering trajectory adversary equipped with a Markov mobility prior (a Viterbi/smoothing adversary is strictly stronger and would only lower the reported trajectory privacy). The two adversaries induce different rankings, so a single privacy scale is invalid.
2) MIRAGE, a road-graph-native optimal obfuscation mechanism that scales to a real city (Section 6). MIRAGE solves the optimal-mechanism LP with graph shortest-path distortion and on-graph output support, made tractable by density-adaptive local LPs with an $O(|V|C^3)$ precomputation, whose gap to the global optimum we measure directly (within 1.5% in the practical regime). DP and k-anonymity arise as constrained or degenerate special cases; an optional geo-indistinguishability constraint recovers a formal ε-DP guarantee.
3) The price of heuristics on real road networks (Sections 8–9). Across three datasets and two cities we quantify, at matched utility, how much privacy DP and k-anonymity sacrifice (5–22% in the practical regime), with 95% bootstrap confidence intervals, and we show the cross-city variation is consistent with the prior heterogeneity that the optimal mechanism exploits.
4) A trajectory analysis with a negative result (Section 13). We build a trajectory-aware variant (MIRAGE-T) and show that snapshot-optimality does not imply trajectory-optimality, because the scalability partition leaks region membership. This isolates the central open problem.
5) A reproducible, real-network, multi-city framework and open release spanning a grid abstraction and real OSM graphs of Beijing and Porto, three datasets, ablations, scalability measurements, and a radio-technology sensitivity analysis of the energy model.[1]

For positioning, we do not claim to invent optimal obfuscation, which is due to Shokri et al. [5], [6] and Bordenabe et al. [7]. Our contribution is to make it graph-native, city-scale, and measured on real road networks, to explain when it helps, and to place it within a unified two-threat evaluation.

2 Preliminaries and Notation

We model the environment as an undirected weighted graph $G=(V,E)$ whose vertices are discrete locations (road-network nodes) and whose edge weights are road lengths; $d_G(u,v)$ denotes shortest-path distance in metres. A user is at a true node $v \in V$; a mechanism releases an observation o drawn from a conditional distribution $f(o \mid v)$, the obfuscation mechanism. We restrict outputs to graph nodes, $o \in V$, so every release is a valid on-road location.

1. https://github.com/Pushkal-Gupta/Graph-based-Location-Privacy-IoT

TABLE 1
Notation.

Symbol	Meaning
$G=(V,E)$	road-network graph; d_G shortest-path distance (m)
$\pi(v)$	population prior (occupancy) at node v
$\mathbf{T}$	Markov mobility transition matrix, $\mathbf{T}_{vw}=P(w \mid v)$
$f(o \mid v)$	mechanism: release prob. of o given true v
$D_{\max}$	utility budget: max expected distortion (m)
$\hat{v}$	adversary's Bayes estimate of the true node
C	MIRAGE local-region size
ε	(geo-ind) privacy budget (per metre)

The adversary and mechanism share public background knowledge: a population prior $\pi(v)$, the marginal probability that an arbitrary report originates at v (estimated as normalised node occupancy), and a first-order Markov mobility model $P(v_{t+1} \mid v_t)$ with transition matrix $\mathbf{T}$, which captures how users move between successive time steps. Table 1 summarises the notation.

Utility. We measure utility loss as expected distortion $\mathrm{U}(f)=\sum_v \pi(v)\sum_o f(o \mid v)\, d_G(v,o)$, the mean displacement of released from true locations, which is the quantity that analytics degrade with.

Privacy. Following Shokri et al. [4], privacy is the expected error of an optimal adversary who inverts f using π (and, for trajectories, $\mathbf{T}$). It is measured in metres and is identical in form for every mechanism, so it is directly comparable, unlike k or ε.

3 Related Work

Heuristic mechanisms. Spatial cloaking and k-anonymity generalise a location to a region containing $\geq k$ users [1], [2], [10], [11]; geo-indistinguishability adds planar Laplace noise with a metric privacy guarantee [8], [9]; graph projection snaps noisy points to the road network [7]; density-aware schemes vary k locally [12]; temporal cloaking delays reports until k users co-occur [2]. All are heuristics: they fix a rule and accept whatever privacy it yields, and are typically evaluated within one family.

Optimal mechanisms. Shokri et al. [5] showed that the utility-optimal mechanism against a Bayesian localisation adversary is an LP; [6] casts it as a Stackelberg game; Bordenabe et al. [7] add geo-indistinguishability as linear constraints. All of this work is on the Euclidean plane or regular grids, is snapshot-focused, and is explicitly scalability-limited (the LP has $|V|^2$ variables, about 10^7 for our graphs); none is evaluated on a real road network. MIRAGE closes exactly this gap and additionally measures the resulting benefit at city scale.

Trajectory correlation. Xiao and Xiong [13] showed that per-report DP degrades under Markov temporal correlation; subsequent work handles trajectories via DP composition, dummy trajectories, or synthesis [14]–[16]. We instead use a causal Bayesian filtering trajectory adversary to measure this degradation across mechanisms on a common scale, and study whether an optimal snapshot mechanism is also trajectory-robust.

Quantifying privacy. Shokri et al. [4] define location privacy as the expected error of an optimal adversary, the

TABLE 2
Positioning vs. closest prior work. G=graph-native metric, Sc=city-scale, RN=evaluated on a real road network, Tr=trajectory adversary measured, MC=multi-city.

	G	Sc	RN	Tr	MC
Geo-ind (Andrés'13) [8]	no	–	no	no	no
Optimal mech. (Shokri'12) [5]	no	no	no	no	no
Optimal geo-ind (Bordenabe'14) [7]	no	no	no	no	no
DP under correlation (Xiao'15) [13]	no	–	no	yes	no
MIRAGE (this work)	yes	yes	yes	yes	yes

TABLE 3
Datasets after map-matching to the real graphs.

Dataset	City	Points	Graph nodes
GeoLife	Beijing	∼11.4 M	3,154
T-Drive	Beijing	∼4 M	3,154
Porto (taxi)	Porto	∼81 M	3,455

metric we adopt. Table 2 positions this paper against the closest prior work along the axes that matter for deployment.

4 System and Threat Models

Graphs. We use two graphs over central Beijing: a controlled 30×30 grid (900 nodes) that removes topological confounds, and the real OpenStreetMap (OSM) drive network for the same bounding box, consolidated to 3,154 nodes and 5,111 edges. Unlike the grid's uniform degree 4, the real graph has irregular degree (mean 3.24, max 23), 16% dead-ends, and road hierarchy. For a second city we build a real Porto OSM graph (3,455 nodes, mean degree 3.04, 27% dead-ends). Intersections are consolidated within a tolerance so the graph is pipeline-friendly while preserving real topology; shortest-path distances are shared by every mechanism so none is advantaged by a different metric.

Data. Three real mobility datasets across the two cities (Table 3). GeoLife [17], [18] is 182 users of multi-modal GPS in Beijing; T-Drive is roughly 10k Beijing taxis over a week, a much denser regime; the Porto ECML/PKDD taxi dataset is 1.71 M trips. Each is map-matched to the corresponding real graph by projecting GPS to UTM and snapping to the nearest node. Using datasets of very different density, and a second city, separates mechanism properties from single-dataset or single-city quirks.

Two adversaries. Both know the mechanism f and the public knowledge $(\pi,\mathbf{T})$, but not the mechanism's private randomness. A snapshot adversary observes one release; a trajectory adversary observes a sequence and decodes it. We formalise both next.

5 Adversary-Grounded Privacy Metric

5.1 Snapshot Adversary

Observing a release o, the adversary forms the Bayes posterior

$$P(v\mid o)=\frac{f(o\mid v)\,\pi(v)}{\sum_{v'}f(o\mid v')\,\pi(v')}, \tag{1}$$

and outputs the estimate minimising expected graph distance, $\hat{v}(o)=\arg\min_{h\in V}\sum_v P(v\mid o)\,d_G(h,v)$. The privacy of f is the expected error of this optimal estimator,

$$\mathrm{Priv}_{\mathrm{snap}}(f)=\mathbb{E}_{v\sim\pi,\,o\sim f(\cdot|v)}\left[d_G(v,\hat{v}(o))\right]. \tag{2}$$

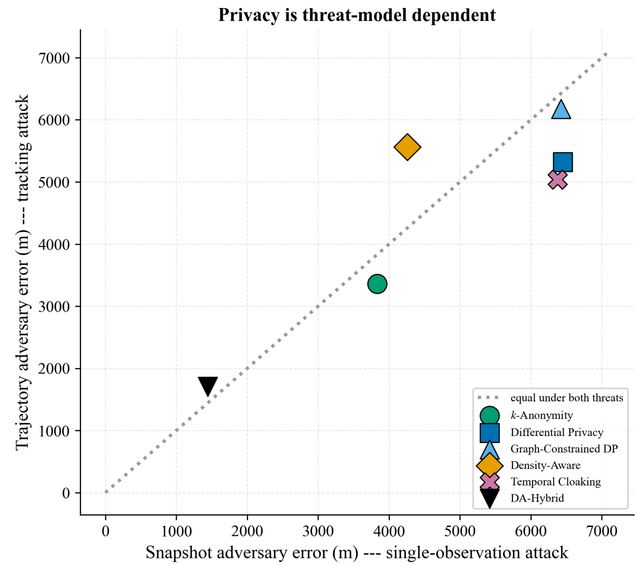

Fig. 1. Privacy is threat-model dependent: mechanisms rank differently under the snapshot (x) and trajectory (y) adversaries; points off the diagonal are protected asymmetrically.

The observation model $f(o\mid v)$ is mechanism-specific: an anonymity set for k-anonymity, a Laplace kernel for DP, the LP solution for MIRAGE. The measured quantity, expected localisation error in metres, is identical for all of them.

5.2 Trajectory Adversary

Given a sequence $o_{1:T}$ for one user, the adversary maintains a belief over the hidden true node using the mobility model. In filtering (causal) form, the belief b_t predicts through mobility and updates through the emission:

$$b_t^-(v)\propto\sum_{v'}b_{t-1}(v')\,\mathbf{T}_{v'v},\quad b_t(v)\propto b_t^-(v)\,f(o_t\mid v). \tag{3}$$

Its per-step estimate is $\hat{v}_t=\arg\min_h\sum_v b_t(v)d_G(h,v)$; the trajectory privacy is the mean tracking error $\frac{1}{T}\sum_t d_G(v_t,\hat{v}_t)$. We report this causal filtering adversary throughout, matched to an online attacker; a Viterbi/smoothing adversary that also uses future observations is strictly stronger and would only lower the reported trajectory privacy for every mechanism. This is the threat that temporal mechanisms target and that per-report noise fails to defeat, because temporal correlation lets the adversary average noise out [13].

5.3 Why a Single Scale Is Inadequate

Fig. 1 plots the two errors against each other for the classical mechanisms on the grid. Density-aware adaptation and temporal cloaking swap rank between the snapshot and trajectory adversaries: a mechanism strong against one is weak against the other. Any single-scale privacy comparison therefore mis-ranks mechanisms that defend different threats. This motivates both the two-threat metric and, later, a trajectory analysis of MIRAGE itself.

6 MIRAGE: Optimal Graph-Native Obfuscation

6.1 The Optimal Mechanism on a Graph

For a prior π and a utility budget $D_{\max}$, we seek the release distribution f that maximises the snapshot adversary's ex-

Algorithm 1 MIRAGE precompute (per graph, offline)

Require: graph G, prior π, budget $D_{\max}$, region size C

partition V into local regions (greedy, densest-first)
for each region R **do**
  $d_R \leftarrow$ graph distances on R; $\pi_R \leftarrow \pi|_R$ (renorm.)
  $f_R \leftarrow$ solve LP (4) on $(d_R, \pi_R, D_{\max})$
  store $f_R(\cdot \mid v)$ for each $v \in R$
end for

pected error (2) subject to the utility budget. The adversary's expected error, given observation o, is a minimisation over its estimate h; introducing one auxiliary variable y_o per observation to encode that inner minimum linearises the problem [5], which gives the LP

$$\max_{f \geq 0, y} \sum_o y_o \qquad \text{(privacy, metres)} \tag{4a}$$

$$\text{s.t.} \quad y_o \leq \sum_v \pi(v) f(o \mid v) \, d_G(h, v) \quad \forall o, h \qquad \text{(adversary)} \tag{4b}$$

$$\sum_v \pi(v) \sum_o f(o \mid v) \, d_G(v, o) \leq D_{\max} \qquad \text{(utility)} \tag{4c}$$

$$\sum_o f(o \mid v) = 1 \quad \forall v. \qquad \text{(stochastic)} \tag{4d}$$

At optimality each y_o equals $\min_h \sum_v \pi(v) f(o \mid v) d_G(h, v)$, so $\sum_o y_o$ is exactly $\mathrm{Priv}_{\mathrm{snap}}(f)$; maximising it yields the privacy-optimal mechanism for the budget. Distances d_G are graph shortest-paths and outputs are graph nodes, so the mechanism is native to the road network, which is new relative to the Euclidean and grid formulations of [5], [7]. An optional geo-indistinguishability constraint

$$f(o \mid v) \leq e^{\varepsilon d_G(v, v')} f(o \mid v') \quad \forall o, v, v' \tag{5}$$

adds a formal ε-DP guarantee, recovering [7] as a constrained special case; k-anonymity (uniform release over the k nearest) and plain geo-ind DP are likewise expressible, so MIRAGE subsumes the heuristics.

6.2 Scaling to a City via Density-Adaptive Local LPs

Program (4) has $|V|^2$ variables and $|V|^2$ adversary constraints (4b), about 10^7 of each for our graphs, and is intractable directly, which is exactly why prior work stayed on small grids. MIRAGE decomposes the graph into density-adaptive local regions and solves one small LP per region (Algorithm 1). Starting from the highest-prior unassigned node, each region greedily takes its C nearest still-unassigned nodes, so dense areas are seeded first and covered by tighter regions. Within a region of size C the LP has $O(C^2)$ variables and $O(C^2)$ constraints and solves in milliseconds; there are $\lceil |V|/C \rceil$ regions, for an $O(|V|C^3)$ offline precomputation (measured in Section 10), after which release is an $O(1)$ categorical draw.

Why we expect the decomposition to be near-lossless. We do not claim a general optimality guarantee for the decomposition; instead we give the intuition and validate the gap empirically. Releasing far from the true node tends to be wasteful under a finite distortion budget, spending utility without buying proportional privacy, because the adversary's error saturates at the prior's spread. An optimal global mechanism should therefore keep $f(o \mid v)$ supported near v, and partitioning V into local regions of radius comparable to the budget preserves most of that support. Two measurements test this: enlarging C raises privacy only marginally beyond the knee (Section 10), and a direct comparison against the global $|V|^2$ LP on a solvable graph bounds the loss at a mean 1.5% for distortion $\leq 500\,\mathrm{m}$ (Fig. 5). A formal upper bound on the gap as a function of region diameter, prior concentration, and budget is left as future work. The one side effect of the partition, revealing region membership, matters only for the trajectory adversary and is analysed in Section 13.

6.3 Trajectory-Aware Variant (MIRAGE-T)

The snapshot LP optimises against the static prior π. Against the trajectory adversary, however, the effective prior at step t is the mobility-propagated belief b_t^- of (3), which is far more concentrated. MIRAGE-T solves, at each step, the region LP with prior $b_t^-|_R$ instead of π_R (receding-horizon, certainty-equivalent optimal control): it hides where the adversary is currently confident. We evaluate whether this closes the gap in Section 13.

7 Experimental Setup

All mechanisms share the graph, data, prior, and distance layer, and we evaluate in two complementary ways. First, the frontier: for each mechanism we compute privacy (optimal-adversary error) and utility (expected distortion) analytically on the same regions, sweeping the mechanism's knob ($D_{\max}$ for MIRAGE, ε for DP, k for k-anonymity), with 95% weighted-bootstrap confidence intervals over regions. This isolates mechanism quality from sampling noise and matches the LP objective. Second, a deployment comparison: all seven mechanisms (the five classical ones, the density-adaptive heuristic DA-Hybrid, and MIRAGE) at representative operating points, under the sampled snapshot and trajectory adversaries, reporting availability and spatial error. The energy model is swept across radio technologies (Section 14). Unless noted, MIRAGE uses C=28. LPs are solved with the HiGHS solver; the shared distance matrix (one dense $|V| \times |V|$ array) lets the exact Bayesian adversary run at roughly 3 k nodes.

Baseline construction and utility matching. Every baseline is instantiated on the same graph, prior, and d_G, and is matched to MIRAGE on expected distortion (the x-axis of the frontier), so no mechanism is compared at a different utility. DP / geo-indistinguishability draws planar Laplace noise of scale $1/\varepsilon$ and projects the noisy point to the nearest graph node, which makes the release on-graph; the projection is a deterministic post-processing step and so preserves the ε-metric guarantee. We sweep ε and report each point at its realised expected distortion. Graph-constrained DP applies the same exponential kernel directly over d_G (release probability $\propto e^{-\varepsilon d_G(v,o)/2}$ over on-graph candidates) rather than in the plane. k-anonymity releases uniformly over the k nodes nearest the true node in d_G (the generalisation set), with k swept; if fewer than k candidate nodes are reachable within the operating radius the user is unserved, which is the source of its availability shortfall on sparse data. Density-aware varies k with local node density, and temporal cloaking delays a report until k users co-occur. All baselines see the same public prior π as MIRAGE; DP and k-anonymity simply do not use it in forming the release, which is the property the price-of-heuristics measurement isolates.

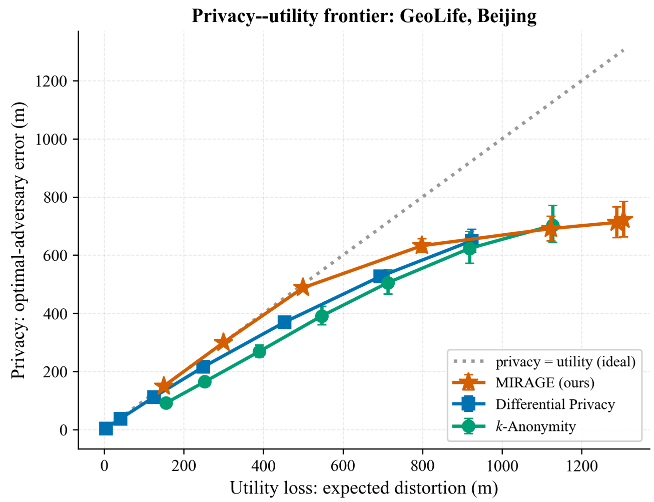

Fig. 2. Privacy–utility frontier (GeoLife, real graph, 95% CI). MIRAGE (near-optimal; within 1.5% of the global LP) dominates the heuristics; the vertical gap to MIRAGE is the price of heuristics. All converge at the intrinsic uncertainty ceiling.

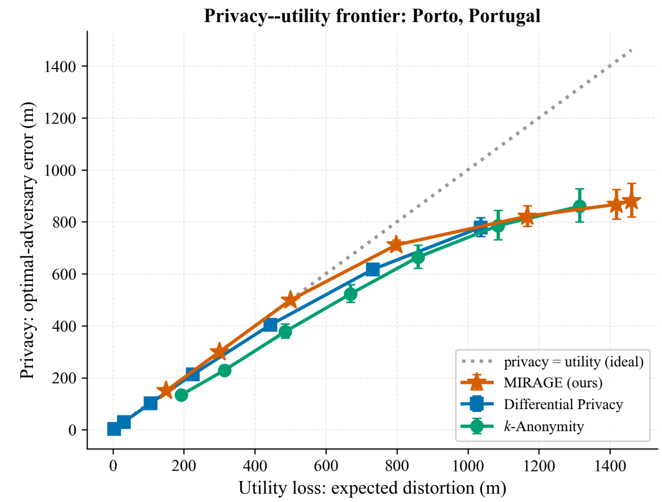

Fig. 3. Cross-city check on the real Porto (Portugal) network under the taxi dataset. MIRAGE dominates here too (+5–12%).

TABLE 4
Price of heuristics (GeoLife, real graph): privacy (optimal-adversary error, m) at matched utility loss.

Utility (m)	MIRAGE	DP	k-anon	MIRAGE gain
150	150	133	92	+13%
300	300	254	201	+18%
500	488	400	355	+22%
800	632	583	555	+9%
≥1100	700+	649	700+	∼ 0% (saturated)

8 Results I: The Price of Heuristics

Fig. 2 shows the privacy–utility frontier on the real GeoLife graph. MIRAGE hugs the ideal line (privacy = utility) at low distortion and dominates DP and k-anonymity throughout the practical regime; all mechanisms converge at high distortion to the network's intrinsic prior-uncertainty ceiling (about 720 m, the maximum error any mechanism can induce given the prior). Table 4 quantifies the gap: at matched utility MIRAGE delivers +13 to +22% more privacy than the best heuristic in the 150–500 m regime on GeoLife, so that fraction of their distortion is wasted by DP and k-anonymity. The gap closes only past about 1100 m, where all mechanisms saturate. A region-level paired bootstrap (113 regions) confirms that the advantage is statistically significant rather than confidence-interval overlap: at matched utility (about 500 m) MIRAGE is more private than DP by +118 m (95% CI $[+109, +128]$, one-sided $p < 0.001$) and than k-anonymity by +97 m ($[+65, +126]$, $p < 0.001$).

Cross-city generalisation. The result holds on all three datasets and both cities (Fig. 3): the practical-regime gain is +13–22% on GeoLife, +9–17% on T-Drive, and +5–12% on Porto. MIRAGE dominates the heuristics everywhere; only the magnitude varies, which we examine next.

9 Results II: Why the Gap Varies

MIRAGE exploits the prior, whereas DP and k-anonymity essentially ignore it, as they hide near-uniformly over neighbours. Its advantage should therefore track how non-uniform occupancy is within a local region: where the prior is skewed there is structure to exploit, and where it is flat the heuristics are already near optimal. Table 5 is consistent with this. The within-region prior entropy (normalised) orders the three cities monotonically with the measured gain: GeoLife is the most concentrated (0.79, gain 20%), Porto the most uniform (0.87, gain 10%), and T-Drive lies between (0.84, 15%). Mobility predictability does not track the gap, as expected, since the snapshot frontier depends on the prior and not on transitions. Fig. 4 plots the relationship. With only three cities this is a correlation rather than an established law, but it suggests an operator rule of thumb: the more spatially concentrated the population, the more an optimal mechanism buys, and the effect is worth measuring on a deployer's own occupancy data.

TABLE 5
Cross-city gap is consistent with prior heterogeneity (n=3): lower within-region entropy (more skewed occupancy) coincides with larger MIRAGE gain.

City	Prior ent.	Within-reg. ent.	Mob. cond. ent.	Gain
GeoLife	0.80	0.79	0.30	20%
T-Drive	0.89	0.84	0.75	15%
Porto	0.74	0.87	0.19	10%

10 Results III: Ablation and Scalability

Table 6 ablates MIRAGE's two design choices. The region size C trades privacy for compute: privacy rises with C (at $D_{\max}$=800, from 421 m at C=12 to 715 m at C=40) while the whole-graph precompute grows as $O(|V|C^3)$ (from 1.3 s to 20.8 s); the knee is at C=28, and the marginal privacy above it confirms that the local decomposition loses little. The geo-indistinguishability constraint (5) costs some optimality ($632 \to 456$ m at ε/m=0.01) in exchange for a formal DP guarantee, which quantifies the price of that guarantee within one framework. All precomputes complete in seconds to tens of seconds on a laptop for a ∼3 k node city, and release is $O(1)$, so MIRAGE is deployable.

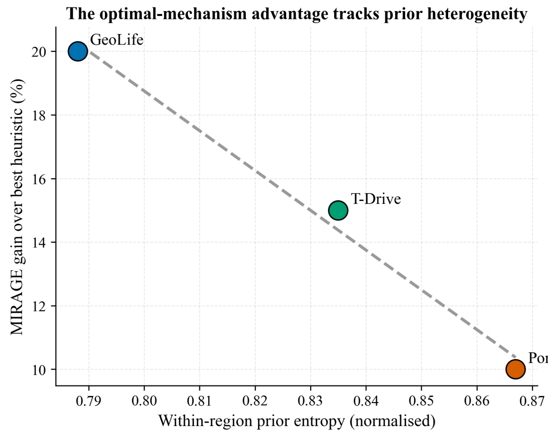

Fig. 4. The optimal-mechanism advantage tracks within-region prior heterogeneity across cities: more concentrated occupancy, larger gain.

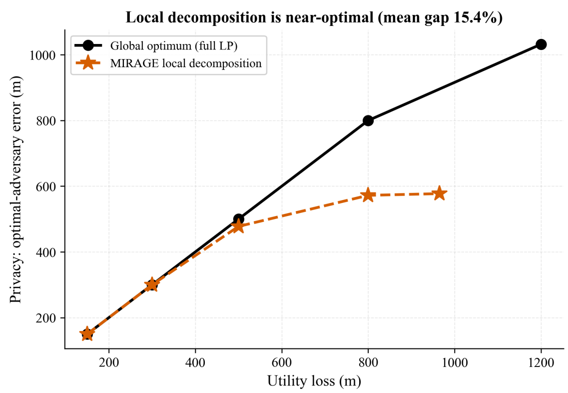

Fig. 5. The scalable local decomposition matches the global LP optimum in the practical regime (mean gap 1.5% for distortion ≤ 500 m), diverging only at high distortion where all mechanisms saturate.

The cost of the decomposition. We measure it exactly on a small graph where the global $|V|^2$ LP is solvable: we compute the true global optimum (one LP over all nodes) and the local decomposition, both scored by the exact optimal-adversary metric (Fig. 5). In the operationally relevant regime the decomposition is essentially lossless, with a mean privacy gap of 1.5% for distortion ≤ 500 m and 0% up to 300 m, so MIRAGE's measured dominance is genuinely near-globally-optimal. A gap appears only at high distortion (28–44% at $D_{\max} \geq 800$), where the global optimum releases across regions but the partition confines releases; that is the saturated regime where all mechanisms converge to the uncertainty ceiling and the advantage has already vanished.

11 Results IV: Deployment Comparison

Table 7 compares all seven mechanisms on the real GeoLife graph under the sampled snapshot and trajectory adversaries. MIRAGE serves every user (100% availability), is on-graph, and provides strong balanced privacy; the two adversaries again reorder the heuristics. Repeating on T-Drive and Porto yields two robustness findings. First, rankings under the adversary metric hold across topology and city. Second, availability is dataset-density-driven, not mechanism-intrinsic: on

TABLE 6
MIRAGE ablation (GeoLife, real graph). Privacy at $D_{\max}$=800 m and whole-graph precompute time vs. region size C.

Region size C	12	20	28	40
Regions	263	158	113	79
Privacy (m)	421	551	632	715
Precompute (s)	1.3	3.1	7.0	20.8

TABLE 7
Deployment comparison (GeoLife, real graph, 10 min). Privacy = Bayesian- adversary error (m, higher = more private).

Mechanism	Snap. AE	Traj. AE	Avail.	Loc. Err
k-Anonymity	1694	3459	33%	4493
Differential Privacy	730	947	100%	466
Graph-Constr. DP	1339	732	100%	1400
Density-Aware	1437	3459	61%	2379
DA-Hybrid (heuristic)	311	1885	100%	563
Temporal Cloaking	2577	4636	100%	3215
MIRAGE (ours)	846	855	100%	1090

dense T-Drive every mechanism reaches 100% availability, whereas on sparse GeoLife k-anonymity (33%) and density-aware (61%) suffer denial of service (Fig. 6). A mechanism's availability is therefore a property of the deployment as much as of the algorithm.

MIRAGE's own deployment point is consistent across all three settings (Table 8): 100% availability, on-graph output, and balanced snapshot/trajectory privacy, with error tracking each network's granularity.

12 Results V: Robustness to a Misspecified Prior

MIRAGE exploits the prior $\boldsymbol{\pi}$, which a deployer must estimate from finite, possibly stale data. To test whether its advantage survives an imperfect estimate, we solve MIRAGE with a corrupted prior $\hat{\boldsymbol{\pi}} = (1-\alpha)\boldsymbol{\pi} + \alpha\mathbf{u}$ ($\mathbf{u}$ uniform, α the degree of ignorance: 0 perfect, 1 no prior at all) while the adversary attacks with the true $\boldsymbol{\pi}$ (Fig. 7). MIRAGE remains the most private for $\alpha \lesssim 0.25$, and a deployer estimating occupancy from the same data operates far below this crossover. Under severe misspecification ($\alpha \geq 0.5$), however, MIRAGE can fall below the prior-free heuristics, because it optimises for the wrong distribution while the adversary uses the right one. This is a precise deployment caveat: the optimal mechanism is only as good as its prior, which motivates a distributionally-robust variant (optimising the worst case over a prior-uncertainty set) as future work.

13 Results VI: Trajectory Adversary and a Negative Result

MIRAGE is near-optimal against the snapshot adversary. Whether that carries to the trajectory adversary is a separate question, and the answer is no, for an instructive structural reason. Using the causal filtering adversary (3) on real user traces (Fig. 8), DP is the most trajectory-private mechanism at matched distortion: at 800 m distortion the tracking error is 676 m for DP versus about 578 m for both MIRAGE variants. The cause is MIRAGE's density-adaptive partition, the very device that makes it scalable and snapshot-optimal.

TABLE 8
MIRAGE deployment point across cities/datasets (real graphs, 10 min, $D_{\max}$=800). Privacy = Bayesian-adversary error (m).

Dataset	Snap. AE	Traj. AE	Avail.	Loc. Err
GeoLife (Beijing)	846	855	100%	1090
T-Drive (Beijing)	645	481	100%	798
Porto (Portugal)	699	1904	100%	798

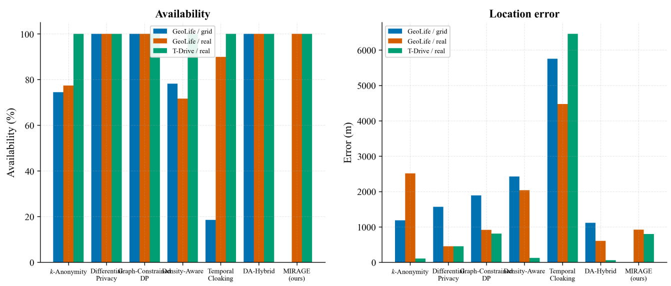

Fig. 6. Availability and error across the grid, the real graph (GeoLife), and the real graph under T-Drive. Availability rises to 100% for all mechanisms on the dense dataset, so it is density-driven.

Because each release lies in a fixed region, observing it reveals region membership, which the sequential adversary exploits to localise; DP's diffuse, unpartitioned noise leaks less across time.

Our trajectory-aware variant MIRAGE-T (per-step LP against the mobility-propagated belief) helps, and its benefit tracks mobility. On stationary GeoLife, where users linger, so the belief barely moves and the dynamic prior is close to the static one, MIRAGE-T essentially matches snapshot-MIRAGE (about 578 vs. 577 m at 800 m distortion). On the mobile T-Drive taxis (low self-transition), where the belief shifts each step and there is real structure to exploit, MIRAGE-T improves trajectory privacy by +9 to +13% over snapshot-MIRAGE at matched utility and nearly closes the gap to DP (624 vs. 647 m at 800 m). So belief-aware release helps most exactly where trajectory tracking is most dangerous, but it does not fully overcome the region-membership leak. This is a clean instance of the paper's thesis, that privacy is threat-model dependent, here even for an optimal mechanism, and it isolates the central open problem: a scalable graph mechanism jointly optimal against snapshot and trajectory adversaries, plausibly via soft or overlapping regions whose releases do not reveal membership.

Removing the membership leak: a structural tension. We tested the natural fix directly: replace the single hard partition with a random mixture of two overlapping partitions, so that a release can originate from either partition's region and no longer reveals one region (Fig. 9). It helps only marginally, by +2 to +7% on the mobile T-Drive traces, and slightly hurts on the stationary GeoLife ones; in neither case does it close the gap to DP (at about 800 m distortion, 527–564 m for MIRAGE variants vs. 669 m for DP). The lesson is that the tension is structural: light de-partitioning is insufficient, because the same local region structure that yields snapshot-optimality inherently concentrates releases. Matching DP's trajectory-robustness appears to require genuinely diffuse support, which sacrifices snapshot optimality. A mechanism that is jointly optimal for both threats, perhaps a many-partition ensemble or a continuous-support graph LP, remains the central open problem that this analysis sharpens.

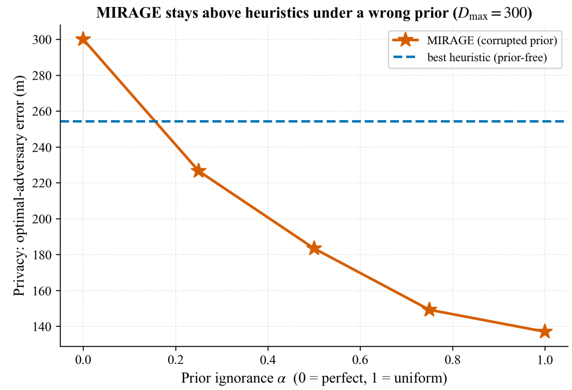

Fig. 7. Robustness to prior misspecification. MIRAGE stays above the best prior-free heuristic while the prior is reasonably accurate ($\alpha \lesssim 0.25$); a badly wrong prior can make it underperform.

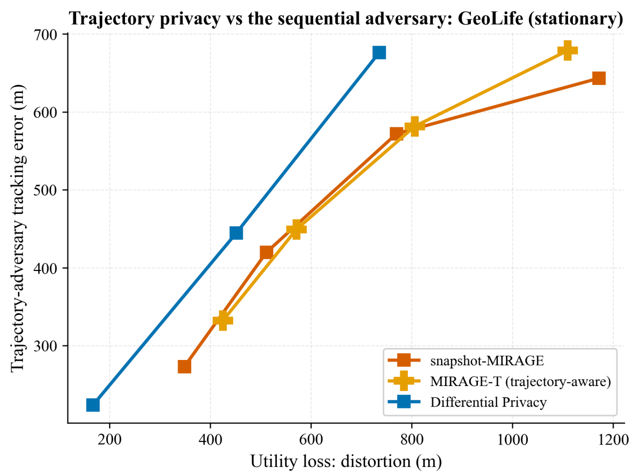

Fig. 8. Trajectory adversary (GeoLife, real traces). At matched distortion DP is most trajectory-private; MIRAGE's region partition leaks membership, and the mobility-aware MIRAGE-T does not close the gap.

14 Results VII: Energy-Model Sensitivity

Per-report energy is $E = E_{\text{radio}} + E_{\text{comp}} + E_{\text{retrans}}$, with $E_{\text{radio}} = P_{\text{tx}} t_{\text{tx}}$. Rather than fix one radio, we sweep BLE, Zigbee, LoRa, NB-IoT, and LTE-M (Fig. 10). The common claim that radio transmission dominates holds only for LPWAN and cellular radios (99% radio share for LoRa, NB-IoT, and LTE-M) but not for BLE, where computation is 6–50% of energy for BFS-based mechanisms. MIRAGE's release-time cost is a single categorical draw ($O(1)$), so it is as radio-dominated as DP; its LP cost is paid once, offline.

15 Discussion

Deployment guidance. For a chosen utility budget on the operator's own road network and mobility statistics, MIRAGE gives the release policy that maximises snapshot-attacker error within each local region and stays within

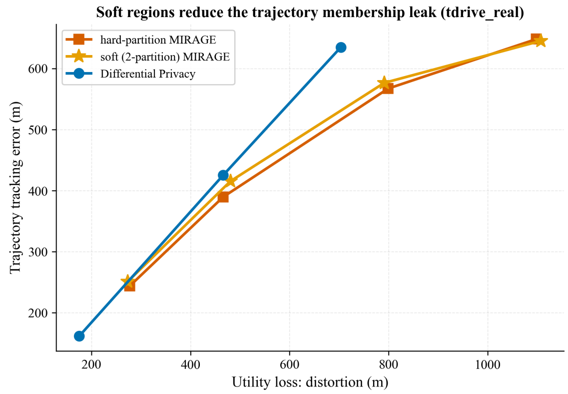

Fig. 9. Attempting to close the trajectory gap with soft (two-partition) regions. It helps marginally on mobile data and not at all on stationary data; DP remains more trajectory-robust, so the membership-leak tension is structural.

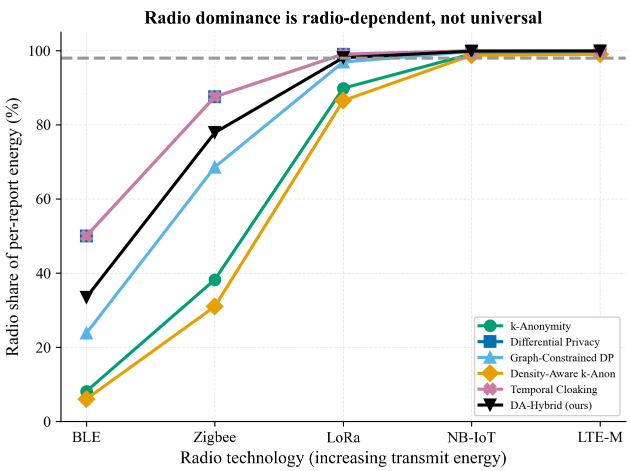

Fig. 10. Radio share of per-report energy vs. radio technology. Radio dominance is radio-dependent, not universal.

1.5% of the global optimum in the practical regime, up to +22% more privacy than tuned heuristics at equal utility, with 100% availability and on-graph outputs, and the gain is largest where the population is spatially concentrated (Section 9). Where a formal DP guarantee is required, the geo-indistinguishability-constrained variant applies at a measured cost. Where sequential tracking is the dominant threat, a diffuse mechanism (or the open membership-hiding design of Section 13) is preferable. The choice must be made against the relevant adversary, which our two-threat metric makes explicit.

Threats to validity. (i) Absolute error is graph-granularity dependent, so cross-topology conclusions rest on adversary-privacy and availability rather than raw error. (ii) The energy model is analytical, suited to ranking rather than absolute prediction; we report its radio sensitivity explicitly. (iii) The mobility prior is first-order Markov; a richer prior would strengthen the trajectory adversary and can only lower reported trajectory privacy, so our trajectory numbers are, if anything, optimistic for all mechanisms. (iv) The local decomposition caps MIRAGE's reach at high distortion, where heuristics catch up at the uncertainty ceiling; hierarchical regions would extend the advantage. (v) MIRAGE's optimality is prior-conditioned (Section 12): a badly misspecified prior can erase its advantage, so a deployer must estimate occupancy reasonably well or use a distributionally-robust variant.

Ethical use. The framework is defensive: it helps operators release less information for a given utility, and it exposes when a mechanism is weaker than its nominal parameter suggests. The adversary models are standard and use only public background knowledge.

16 Conclusion and Future Work

We measured location privacy on a common, adversary-grounded, threat-model-aware scale, and brought optimal obfuscation to a real road network at city scale with MIRAGE. On real networks of two cities across three datasets, deployed heuristics leave 5–22% of achievable privacy on the table in the practical regime, while MIRAGE tracks the global frontier with full availability and on-graph outputs; the advantage is consistent with prior heterogeneity, rankings hold from grid to real topology, and availability is shown to be dataset-density-driven. We also reported a negative result: snapshot-optimality does not imply trajectory-optimality, because the scalability partition leaks region membership. The central open problem, a scalable graph mechanism jointly optimal against snapshot and trajectory adversaries via membership-hiding soft or overlapping regions, together with richer mobility priors and hardware-measured energy, defines our future work. The complete framework is released for reproducible extension.

References

[1] L. Sweeney, "*k*-Anonymity: A model for protecting privacy," Int. J. Uncertainty, Fuzziness Knowl.-Based Syst., vol. 10, no. 5, pp. 557–570, 2002.
[2] M. Gruteser and D. Grunwald, "Anonymous usage of location-based services through spatial and temporal cloaking," in Proc. MobiSys, 2003, pp. 31–42.
[3] C. Dwork, "Differential privacy," in Proc. ICALP, LNCS 4052, 2006, pp. 1–12.
[4] R. Shokri, G. Theodorakopoulos, J.-Y. Le Boudec, and J.-P. Hubaux, "Quantifying location privacy," in Proc. IEEE S&P, 2011, pp. 247–262.
[5] R. Shokri, G. Theodorakopoulos, C. Troncoso, J.-P. Hubaux, and J.-Y. Le Boudec, "Protecting location privacy: Optimal strategy against localization attacks," in Proc. ACM CCS, 2012, pp. 617–627.
[6] R. Shokri, "Privacy games: Optimal user-centric data obfuscation," Proc. Privacy Enhancing Technologies, vol. 2015, no. 2, pp. 299–315, 2015.
[7] N. E. Bordenabe, K. Chatzikokolakis, and C. Palamidessi, "Optimal geo-indistinguishable mechanisms for location privacy," in Proc. ACM CCS, 2014, pp. 251–262.
[8] M. E. Andrés, N. E. Bordenabe, K. Chatzikokolakis, and C. Palamidessi, "Geo-indistinguishability: Differential privacy for location-based systems," in Proc. ACM CCS, 2013, pp. 901–914.
[9] K. Chatzikokolakis et al., "Broadening the scope of differential privacy using metrics," in Proc. PETS, 2013, pp. 82–102.
[10] M. F. Mokbel, C.-Y. Chow, and W. G. Aref, "The new Casper: Query processing for location services without compromising privacy," in Proc. VLDB, 2006, pp. 763–774.
[11] M. Duckham and L. Kulik, "A formal model of obfuscation and negotiation for location privacy," in Pervasive Computing, LNCS 3468, 2005, pp. 152–170.
[12] B. Niu, Q. Li, X. Zhu, G. Cao, and H. Li, "Achieving *k*-anonymity in privacy-aware location-based services," in Proc. IEEE INFOCOM, 2014, pp. 754–762.

[13] Y. Xiao and L. Xiong, "Protecting locations with differential privacy under temporal correlations," in Proc. ACM CCS, 2015, pp. 1298–1309.
[14] Y. Du et al., "LDPTrace: Locally differentially private trajectory synthesis," Proc. VLDB Endow., vol. 16, no. 8, pp. 1897–1909, 2023.
[15] Y. Hu et al., "Real-time trajectory synthesis with local differential privacy," in Proc. IEEE ICDE, 2024, pp. 1685–1698.
[16] E. Buchholz et al., "SoK: Can trajectory generation combine privacy and utility?," Proc. PETS, vol. 2024, no. 3, pp. 75–93, 2024.
[17] Y. Zheng, L. Zhang, X. Xie, and W.-Y. Ma, "Mining interesting locations and travel sequences from GPS trajectories," in Proc. WWW, 2009, pp. 791–800.
[18] Y. Zheng, Q. Li, Y. Chen, X. Xie, and W.-Y. Ma, "Understanding mobility based on GPS data," in Proc. UbiComp, 2008, pp. 312–321.